

Hologram Representation via Quadratic Phase Gaussian Splatting

Haolong Wang
Swansea University
Swansea, United Kingdom
haolong.wang.00@gmail.com

Yicheng Zhan
University College London
London, United Kingdom
yicheng.zhan.21@ucl.ac.uk

Kaan Akşit
University College London
London, United Kingdom
kaanaksit@kaanaksit.com

Simeng Qiu
Swansea University
Swansea, United Kingdom
simeng.qiu@swansea.ac.uk

Abstract

We introduce Complex-Valued Quadratic Phase Gaussian (CVQPG), a novel hologram representation method that augments each 2D Gaussian primitive with a quadratic phase profile controlled by a learnable curvature parameter. Against the planar Gaussian baseline, CVQPG improves the average PSNR of holographic reconstructions by 0.19 dB (RGB) and 0.33 dB (grayscale) at equal primitive counts, and by 0.05 dB (RGB) and 0.08 dB (grayscale) at equal parameter counts, where it still leads in all visual quality metrics. Our frequency-domain analysis shows that CVQPG better preserves the mid-to-high frequency band of natural images, where the reconstruction MSE drops by up to 11% (RGB) and 22% (grayscale), indicating that modulating primitive wavefronts is an effective and lightweight enhancement.

CCS Concepts

• **Computing methodologies** → **Computer graphics; Rasterization; Computer vision representations.**

ACM Reference Format:

Haolong Wang, Yicheng Zhan, Kaan Akşit, and Simeng Qiu. 2026. Hologram Representation via Quadratic Phase Gaussian Splatting. In *SIGGRAPH Asia 2026 Technical Communications (SA Technical Communications '26)*, December 01–04, 2026, Kuala Lumpur, Malaysia. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3829339.3847858>

1 Introduction

Computer-Generated Holography (CGH) plays an essential role in applications such as 3D displays [Kavaklı et al. 2023; Zhan et al. 2026a; Zhang et al. 2017]. Given the dense spatial variations in holograms, designing efficient representations that preserve high-frequency details remains a major challenge in CGH. Implicit Neural Representations (INR) [Sitzmann et al. 2020] favor low-frequency content, and Gaussian image representations [Zhang et al. 2024] fit 2D Gaussians to pixels for compression. A separate line of research tackles hologram novel view synthesis using Gaussians [Choi et al. 2025; Zhan et al. 2026b]. Closest to us, Zhan et al. [2026a] place complex-valued Gaussians directly on the 2D hologram plane to model interference and diffraction, enabling efficient and high-fidelity reconstruction.

The work by Zhan et al. [2026a] assumes a flat phase profile for each primitive, leaving the representational capacity of Gaussian primitives an under-explored question. We further explore this topic by representing full-complex holograms using primitives that

explicitly model the curvature of the field. Our proposed Complex-Valued Quadratic Phase Gaussian (CVQPG) augments each Gaussian with a quadratic phase profile controlled by a learnable curvature parameter. We also derive a per-primitive curvature bound from the display’s Nyquist sampling limit, ensuring robust hologram reconstruction without high-frequency aliasing.

Our experiments show that CVQPG achieves consistent quality improvements and outperforms the baseline in parameter effectiveness. We investigate the frequency domain features of CVQPG and ablate its components. Our code is available at [GitHub:gsmark36/complex-valued-quadratic-phase-gaussian](https://github.com/GitHub:gsmark36/complex-valued-quadratic-phase-gaussian).

2 Methods

Problem Definition. The holographic display can reconstruct a 3D scene as multi-plane intensity images by propagating light from a complex hologram $\mathbf{H} \in \mathbb{C}^{N_c \times H \times W}$ with N_c color channels. To synthesize the intensity target for each depth plane, we take an image $\mathbf{I}_{target} \in \mathbb{R}^{N_c \times H \times W}$ and its depth map $\mathbf{D} \in \mathbb{R}^{H \times W}$ as the input for the target synthesis function f_{synth}^m at plane m . The optimization problem is therefore formulated as:

$$\hat{\mathbf{H}} \leftarrow \argmin_{\mathbf{H}} \sum_{m=1}^M \mathcal{L}(f_{recon}^m(\mathbf{H}), f_{synth}^m(\mathbf{I}_{target}, \mathbf{D})), \quad (1)$$

where M denotes the total number of depth planes, f_{recon}^m is the reconstruction function at plane m , and $\mathcal{L}$ represents the loss function that measures the difference between the reconstructed image and the synthesized target.

Complex-Valued Planar Gaussian. Our method builds upon the complex-valued 2D Gaussian representation [Zhan et al. 2026a], which we refer to as the planar Gaussian baseline model. It encodes a hologram as a set of N Gaussian primitives. Each primitive is parameterized as $\mathcal{G} = \{\alpha, \mathbf{c}, \mathbf{x}, \mathbf{s}, \theta, \boldsymbol{\varphi}\}$, where $\alpha \in \mathbb{R}$ denotes the opacity, $\mathbf{c} \in \mathbb{R}^{N_c}$ the per-channel color amplitude, $\mathbf{x} \in \mathbb{R}^2$ the 2D position, $\mathbf{s} \in \mathbb{R}^2$ the scales, $\theta \in \mathbb{R}$ the rotation angle, and $\boldsymbol{\varphi} \in \mathbb{R}^{N_c}$ the per-channel phase. The distribution of the Gaussian primitive is determined by the 2D covariance matrix $\Sigma = \mathbf{R}\mathbf{S}\mathbf{S}^T\mathbf{R}^T$, where $\mathbf{R}$ is the rotation matrix of angle θ , and $\mathbf{S} = \text{diag}(\mathbf{s})$ is the scaling matrix of scales $\mathbf{s}$. Therefore, the contribution of a Gaussian primitive at pixel coordinate $\mathbf{p}$ is

$$\mathcal{G}(\mathbf{p}) = \alpha \cdot \mathbf{c} \cdot \exp(-\frac{1}{2}(\mathbf{p} - \mathbf{x})^T \Sigma^{-1}(\mathbf{p} - \mathbf{x})) \cdot \exp(j\boldsymbol{\varphi}). \quad (2)$$

The phase profile of each primitive is flat across its effective area, equivalent to a plane wave emitted from a Gaussian ellipse.

Complex-Valued Quadratic Phase Gaussian. Considering the representational capacity, we present a novel primitive: Complex-Valued Quadratic Phase Gaussian (CVQPG). Our design incorporates the Quadratic Phase Factor (QPF) to explicitly modulate field



This work is licensed under a Creative Commons Attribution 4.0 International License. *SA Technical Communications '26*, Kuala Lumpur, Malaysia
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2841-9/2026/12
<https://doi.org/10.1145/3829339.3847858>

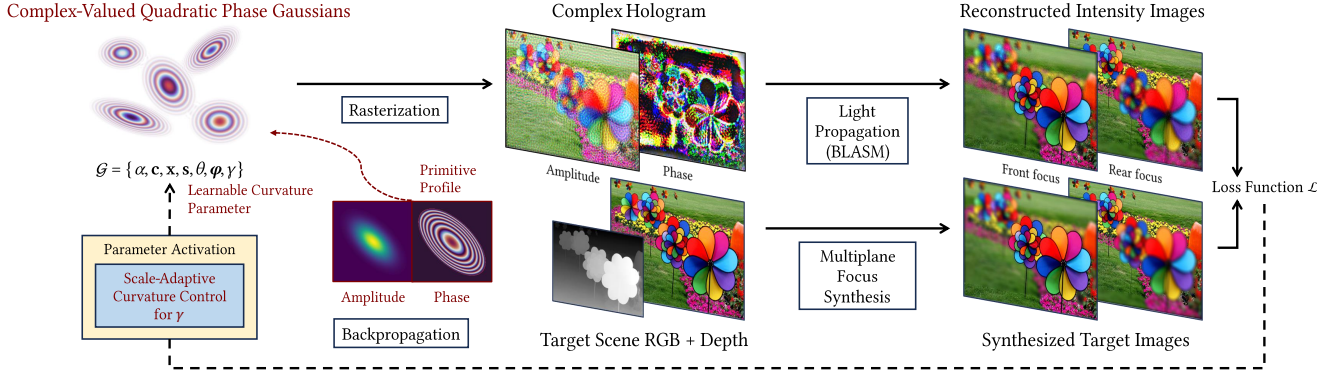


Figure 1: CVQPG represents full complex holograms with an extra learnable curvature parameter γ , and optimizes the hologram by comparing the reconstructed intensity images with the synthesized targets. (Source Image: [Spragg 2013])

curvature with a minimal overhead of one extra parameter. Conventionally, we modulate a Gaussian primitive with a standard QPF,

$$Q_{std}(\mathbf{p}) = \exp(j \frac{k}{2f} \|\mathbf{p} - \mathbf{x}\|^2), \quad (3)$$

where k is the wavenumber, and f is the focal length. However, the standard QPF models an isotropic circular wavefront, leading to a mismatch with the Gaussian's anisotropic amplitude envelope (Fig. 2). To resolve this, we utilize the squared Mahalanobis distance $D_{mah}^2(\mathbf{p}) = (\mathbf{p} - \mathbf{x})^\top \Sigma^{-1}(\mathbf{p} - \mathbf{x})$ as a unified spatial metric. Therefore, the shape-adaptive Mahalanobis QPF is formulated as

$$Q_{mah}(\mathbf{p}) = \exp(j \frac{k\gamma \det(\mathbf{S})}{2} D_{mah}^2(\mathbf{p})), \quad (4)$$

where $\gamma = 1/f$ is the learnable curvature parameter representing optical power, and $\det(\mathbf{S})$ is the determinant of the scaling matrix $\mathbf{S}$. Consequently, each CVQPG primitive is parameterized as $\mathcal{G} = \{\alpha, c, \mathbf{x}, s, \theta, \phi, \gamma\}$, with an added scalar parameter $\gamma \in \mathbb{R}$. Hence, the contribution of a CVQPG primitive at pixel coordinate $\mathbf{p}$ is given by

$$\mathcal{G}(\mathbf{p}) = \alpha \cdot c \cdot \exp(-\frac{1}{2} D_{mah}^2(\mathbf{p})) \cdot \exp(j\phi) \cdot Q_{mah}(\mathbf{p}). \quad (5)$$

The configuration of a CVQPG primitive is equivalent to the combination of a planar Gaussian light source and a virtual lens with optical power γ . During optimization, we symmetrically bound γ within $[-\gamma_{\max}, \gamma_{\max}]$, which enables a continuous wavefront transformation from divergence ($\gamma > 0$) to convergence ($\gamma < 0$). We can represent a planar wavefront by $\gamma = 0$, avoiding numerical instability caused by the focal length $f \rightarrow \pm\infty$.

Scale-Adaptive Curvature Control. The quadratic phase profile acts as a linear chirp signal, providing high-frequency components for detailed hologram reconstruction. However, to prevent aliasing artifacts caused by the physical bandwidth limitations of the Spatial Light Modulator (SLM), we impose a frequency constraint on each primitive. By restricting the chirp evaluation to the 3σ footprint of each primitive, the maximum instantaneous spatial frequency is

$$v_{\max} = \frac{\gamma \cdot 3\sigma \cdot (\Delta x)^2}{\lambda}, \quad (6)$$

where $\sigma = \max(s)$ is the equivalent standard deviation, $\lambda = 2\pi/k$ is the wavelength, and Δx is the pixel pitch of the SLM. Constraining

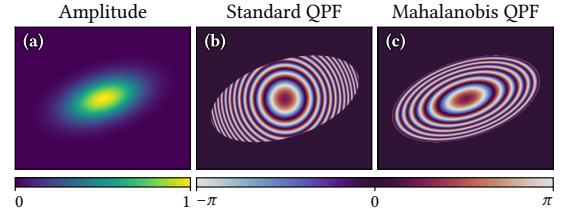


Figure 2: Comparison of QPFs on a Gaussian primitive. (a) Shared amplitude envelope. Unlike (b) the standard QPF with misaligned circular fringes, (c) the Mahalanobis QPF yields elliptical fringes perfectly aligned with the envelope.

this peak frequency below the SLM's Nyquist limit ($v_{\max} \leq 0.5$ cycles/pixel) directly yields the scale-adaptive curvature bound for each primitive:

$$\gamma_{\max} = \eta \cdot \frac{\lambda}{6\sigma \cdot (\Delta x)^2}, \quad (7)$$

where $\eta = 0.9$ is a relaxation factor.

Hologram Rendering and Optimization. In the rendering process, rasterization is applied to accumulate all primitive contributions (Eq. 5) across the pixel grid to form the hologram,

$$\mathbf{H} = \left\{ \sum_{n=1}^N \mathcal{G}_n(\mathbf{p}) \mid \mathbf{p} \in [1, W] \times [1, H] \right\}, \quad (8)$$

where $H \times W$ is the size of the hologram. We propagate the hologram via the conventional Band-limited Angular Spectrum Method (BLASM) to reconstruct the field intensity $I_m = |\text{BLASM}_m(\mathbf{H})|^2$ at plane m . At each plane, we compare the reconstructed intensity I_m with the synthesized target intensity $\hat{I}_m$, minimizing the visual difference between them. The per-plane loss function consists of a reconstruction loss $\mathcal{L}_{recon}$ proposed by Kavaklı et al. [2023] and a similarity loss $\mathcal{L}_{SSIM} = (1 - \text{SSIM})$. To accurately simulate depth-dependent focus and defocus effects, the total training loss $\mathcal{L}$ is summed over all depth planes as

$$\mathcal{L} = \sum_{m=1}^M (\mathcal{L}_{recon}^m(I_m, \hat{I}_m) + w_1 \cdot \mathcal{L}_{SSIM}^m(I_m, \hat{I}_m)), \quad (9)$$

where $w_1 = 0.005$ is the SSIM loss weight.

Curvature Warm-Up Strategy. We optimize the learnable curvature parameter γ via a two-stage strategy. We initialize and lock the curvature at $\gamma = 0$ for all primitives, utilizing the planar Gaussians to optimize the low-frequency patterns. In the refinement stage, the curvature parameter is unlocked and optimized with other parameters. This coarse-to-fine approach stabilizes early-stage training and allows the model to refine the residual high-frequency details.

3 Evaluation and Discussion

Implementation. We utilize a $3.74 \mu\text{m}$ pixel pitch and RGB wavelengths of 639 nm, 532 nm, and 473 nm to propagate the holograms to two depth planes at 1 mm and 5 mm. Shared parameters follow the same initialization schemes as the baseline model [Zhan et al. 2026a]. γ_{max} is calculated from the shortest wavelength, and γ is bounded by a tanh activation. We train the models using identical PyTorch pipelines and the Adan optimizer [Xie et al. 2024]. The training runs for 2000 steps, with a 400-step warm-up stage. All experiments are conducted on a single NVIDIA TITAN RTX GPU. We display the reconstructed holograms on our holographic display prototype (see Supplementary Section 1).

Quantitative Evaluations. The visual quality of reconstructed images is evaluated using five metrics, including Peak Signal-to-Noise Ratio (PSNR), Structural Similarity (SSIM), Perceptual Similarity Metric (LPIPS) [Zhang et al. 2018], FLIP [Andersson et al. 2020], and ColorVideoVDP (CVVDP) [Mantiuk et al. 2024]. Our dataset is derived from both DIV2K [Agustsson and Timofte 2017] and the Real Forward-Facing dataset [Mildenhall et al. 2019]. Then we generate the depth map using Depth Anything V2 [Yang et al. 2024]. We first compare CVQPG against the planar Gaussian baseline on ten scenes at a resolution of 640×480 with compression ratios of 10% and 20%. Under equal primitive counts, CVQPG outperforms the baseline across all metrics, achieving mean PSNR gains of +0.19 dB in RGB and +0.33 dB in grayscale (Tab. 1). Since CVQPG adds one trainable scalar per primitive that brings minor parameter overhead, we further conduct an equal-parameter evaluation by enlarging the baseline’s primitive count by 13/12 $\times$ in RGB and 9/8 $\times$ in grayscale (see Baseline⁺ rows in Tab. 1). Per-scene results are given in Section 3 of the supplementary material. To assess improvements in the frequency domain, we measure the reconstruction MSE at each spatial frequency against the target. Evaluated on ten scenes at an equal primitive count (Fig. 3), the MSE reductions of CVQPG over the baseline are negligible near zero frequency, peak in the mid-to-high frequency band (11% RGB, 22% grayscale), and vanish near the Nyquist limit. Training time and peak GPU memory are reported in Section 2 of the supplementary material.

Qualitative Observations. To further explore the high-frequency feature of CVQPG, we reconstruct a single-plane, grayscale Siemens star. With equal primitives, CVQPG achieves a significant PSNR gain (+3.04 dB) with a distinct visual improvement in the central region (Fig. 5a), demonstrating its capacity to resolve high-frequency details. CVQPG exhibits ringing artifacts under zero-padding due to the abrupt truncation of its chirp phase, which is avoided by the baseline (Fig. 5b). We suppress this by applying a cosine-tapered window to smooth the hologram edges before propagation.

Table 1: Quantitative comparison averaged over ten scenes and two depth planes. We compare CVQPG against the standard Baseline (equal primitive count N) and the enlarged Baseline⁺ (equal parameter count). Δ rows show CVQPG’s improvement over the standard Baseline. Best in bold.

Setting	Method	N	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FLIP $\downarrow$	CVVDP $\uparrow$
RGB 20%	Baseline	30,720	29.265	0.8307	0.2895	0.1144	9.168
	Baseline ⁺	33,280	29.386	0.8336	0.2853	0.1134	9.186
	CVQPG	30,720	29.450	0.8364	0.2828	0.1125	9.203
	Δ		+0.185	+0.0057	-0.0066	-0.0019	+0.035
RGB 10%	Baseline	15,360	28.034	0.7992	0.3292	0.1292	8.917
	Baseline ⁺	16,640	28.179	0.8034	0.3241	0.1271	8.950
	CVQPG	15,360	28.225	0.8053	0.3217	0.1265	8.966
	Δ		+0.191	+0.0060	-0.0076	-0.0026	+0.049
Grayscale 20%	Baseline	15,360	29.504	0.8340	0.3813	0.0753	8.935
	Baseline ⁺	17,280	29.753	0.8394	0.3766	0.0733	8.988
	CVQPG	15,360	29.834	0.8443	0.3694	0.0725	9.019
	Δ		+0.331	+0.0103	-0.0119	-0.0028	+0.085
Grayscale 10%	Baseline	7,680	28.076	0.7964	0.4204	0.0888	8.548
	Baseline ⁺	8,640	28.317	0.8030	0.4134	0.0862	8.621
	CVQPG	7,680	28.398	0.8079	0.4069	0.0850	8.658
	Δ		+0.322	+0.0115	-0.0135	-0.0038	+0.109

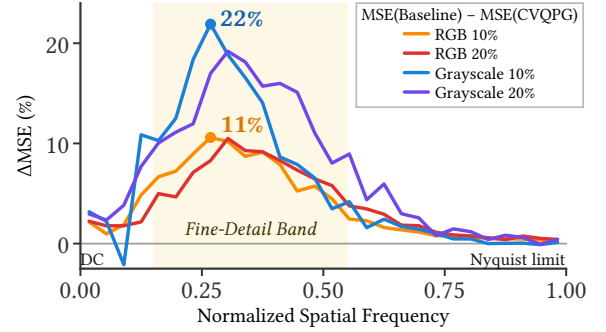


Figure 3: Frequency-resolved Δ MSE of CVQPG over the baseline for ten scenes. The gains cover the mid-to-high frequency band that contains fine details of natural images.

Ablation study. The ablation study covers two components: the Scale-Adaptive Curvature Control and the Mahalanobis QPF. Starting from full CVQPG, we compare five scenes at the RGB 10% budget (Tab. 2). Replacing the scale-adaptive bound with a constant bound significantly reduces the PSNR gain (+0.185 $\rightarrow$ +0.048 dB), identifying it as the decisive component. Replacing the Mahalanobis QPF with the standard QPF costs 0.036 dB on average, which confirms the benefit of the anisotropic phase profile. Finally, a vanilla model without the two components retains only a gain of +0.046 dB. This performance floor indicates that simply appending a standard QPF fails to capture the full potential of phase modulation.

Conclusion and Discussion. CVQPG improves reconstruction quality across all metrics without per-scene tuning and maintains its advantage under equal-parameter evaluation, showing that optimizing wavefronts is more effective than scaling up the count of planar Gaussians. Experiments show that our method achieves its full potential only when the curvature is strictly bounded by the per-primitive sampling limit. We also find that CVQPG preserves

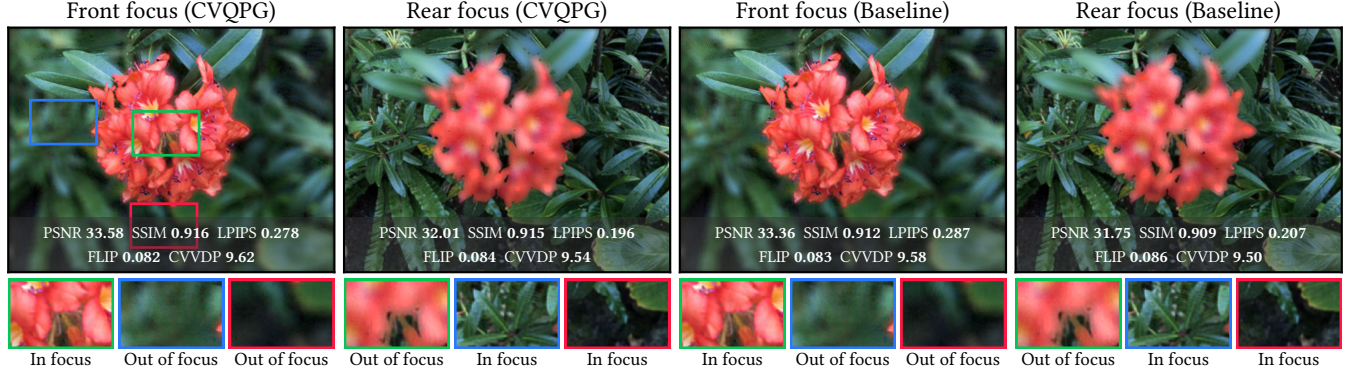


Figure 4: Visual comparison on simulated holographic reconstructions (RGB, 20%, $N=30,720$) at the front (1 mm) and rear (5 mm) depth planes. CVQPG provides improved visual quality over the baseline. (Source Image: [Mildenhall et al. 2019])

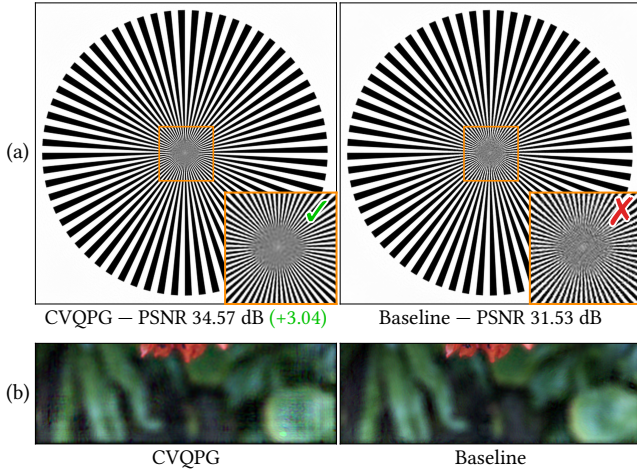


Figure 5: Qualitative comparisons. (a) CVQPG outperforms the baseline by +3.04 dB PSNR on the Siemens star target, providing distinct high-frequency details in the central region. (b) Abrupt boundary truncation induces ringing artifacts in CVQPG, which are absent in the baseline.

Table 2: Component ablation averaged over five scenes (RGB, 10%, $N=15,360$). Δ denotes the mean PSNR gain over the baseline. Best in bold.

Variant	Δ	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FLIP $\downarrow$	CVVDP $\uparrow$
Baseline	—	29.809	0.8419	0.3119	0.1128	9.095
Full (CVQPG)	+0.185	29.994	0.8471	0.3041	0.1109	9.137
w/o Adaptive Bound	+0.048	29.857	0.8438	0.3103	0.1122	9.108
w/o Mahalanobis QPF	+0.149	29.958	0.8462	0.3044	0.1108	9.134
None (Vanilla)	+0.046	29.855	0.8437	0.3098	0.1118	9.109

mid-to-high frequencies of natural images better than the baseline, making the gain content-dependent. Tab. 1 reveals a chromatic limitation: the grayscale gains notably exceed the RGB ones because a shared curvature cannot be optimal for all wavelengths. In future work, we plan to explore wavelength-based curvature to address

the chromatic limitation. We also aim to explore advanced phase formulations and further investigate the representational capacity of explicit hologram representations.

References

- Eirikur Agustsson and Radu Timofte. 2017. NTIRE 2017 Challenge on Single Image Super-Resolution: Dataset and Study. In *2017 IEEE CVPRW*. 1122–1131. doi:10.1109/CVPRW.2017.150
- Pontus Andersson, Jim Nilsson, Tomas Akenine-Möller, Magnus Oskarsson, Kalle Åström, and Mark D. Fairchild. 2020. FLIP: A Difference Evaluator for Alternating Images. *Proc. ACM Comput. Graph. Interact. Tech.* 3, 2, Article 15 (Aug. 2020), 23 pages. doi:10.1145/3406183
- Suyeon Choi, Brian Chao, Jacqueline Yang, Manu Gopakumar, and Gordon Wetzstein. 2025. Gaussian Wave Splatting for Computer-Generated Holography. *ACM TOG* 44, 4, Article 57 (July 2025), 13 pages. doi:10.1145/3731163
- Koray Kavakli, Yuta Itoh, Hakan Ürey, and Kaan Akşit. 2023. Realistic Defocus Blur for Multiplane Computer-Generated Holography. In *2023 IEEE VR*. 418–426. doi:10.1109/VR55154.2023.00057
- Rafal K. Mantiuk, Param Hanji, Maliha Ashraf, Yuta Asano, and Alexandre Chapiro. 2024. ColorVideoVDP: A visual difference predictor for image, video and display distortions. *ACM TOG* 43, 4, Article 129 (July 2024), 20 pages. doi:10.1145/3658144
- Ben Mildenhall, Pratul P. Srinivasan, Rodrigo Ortiz-Cayon, Nima Khademi Kalantari, Ravi Ramamoorthi, Ren Ng, and Abhishek Kar. 2019. Local light field fusion: practical view synthesis with prescriptive sampling guidelines. *ACM TOG* 38, 4, Article 29 (July 2019), 14 pages. doi:10.1145/3306346.3322980
- Vincent Sitzmann, Julien N. P. Martel, Alexander W. Bergman, David B. Lindell, and Gordon Wetzstein. 2020. Implicit Neural Representations with Periodic Activation Functions. arXiv:2006.09661 [cs.CV]
- Bernard Spragg. 2013. Colorful windmills Shenzhen China. Flickr.
- Xingyu Xie, Pan Zhou, Huan Li, Zhouchen Lin, and Shuicheng Yan. 2024. Adan: Adaptive Nesterov Momentum Algorithm for Faster Optimizing Deep Models. *IEEE TPAMI* 46, 12 (2024), 9508–9520. doi:10.1109/TPAMI.2024.3423382
- Lihe Yang, Bingyi Kang, Zilong Huang, Zhen Zhao, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. 2024. Depth Anything V2. arXiv:2406.09414 [cs.CV]
- Yicheng Zhan, Xiangjun Gao, Long Quan, and Kaan Akşit. 2026a. Complex-Valued 2D Gaussian Representation for Computer-Generated Holography. In *Computer Vision – ECCV 2026*. Springer Nature Switzerland, Cham, 557–576. doi:10.1007/978-3-032-37574-2_31
- Yicheng Zhan, Dong-Ha Shin, Seung-Hwan Baek, and Kaan Akşit. 2026b. Complex-Valued Holographic Radiance Fields. *ACM TOG* 45, 3, Article 31 (April 2026), 16 pages. doi:10.1145/3804450
- Jingzhao Zhang, Nicolas Pégard, Jingshan Zhong, Hillel Adesnik, and Laura Waller. 2017. 3D computer-generated holography by non-convex optimization. *Optica* 4, 10 (Oct 2017), 1306–1313. doi:10.1364/OPTICA.4.001306
- Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. 2018. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. In *2018 IEEE/CVF CVPR*. 586–595. doi:10.1109/CVPR.2018.00068
- Xinjie Zhang, Xingtong Ge, Tongda Xu, Dailan He, Yan Wang, Hongwei Qin, Guo Lu, Jing Geng, and Jun Zhang. 2024. GaussianImage: 1000 FPS Image Representation and Compression by 2D Gaussian Splatting. arXiv:2403.08551 [eess.IV]

Hologram Representation via Quadratic Phase Gaussian Splatting

1 Holographic Display Prototype

We employ a holographic display prototype to display reconstructed holograms in our experiments (Fig. 1). A laser source (LASOS MCS4) combines three laser lines and couples them into a single-mode fiber. A plano-convex lens (Thorlabs LA1708-A, $f = 200$ mm) collimates the light from the fiber, and a beamsplitter (Thorlabs BP245B1) reflects the collimated, linearly polarized beam onto a phase-only SLM (Jasper JD7714, 4096×2400 resolution, $3.74 \mu\text{m}$ pixel pitch). The modulated beam propagates through two lenses (Thorlabs LA1908-A and LB1156-A, $f = 500$ mm and $f = 250$ mm), and a pinhole (Thorlabs SM1D12C) in their common focal plane performs spatial filtering. A lensless image sensor (Point Grey GS3-U3-23S6M-C) mounted on an X-stage (Thorlabs PT1/M) captures the reconstructed holographic images. We further fit CVQPG at 3840×2160 , encode the holograms into phase-only patterns, and demonstrate them on our display prototype. Fig. 2 shows an example of the experimentally captured result.

2 Runtime and Memory Consumption

We profile the training runtime and peak memory consumption of the PyTorch pipelines. CVQPG trains in 8.7–26.5 min using 5.5–18.9 GiB of memory, adding at most 3.4% time and 5.9% memory overhead over the Baseline at equal primitive counts (Tab. 1). Under equal-parameter evaluation, CVQPG trains 3.4–5.4% faster and uses 3.8–6.8% less memory than Baseline⁺ while achieving higher reconstruction quality.

Table 1: Training time and peak GPU memory averaged over the ten scenes (PyTorch pipeline, 2000 steps, 640×480 , two depth planes, a single NVIDIA TITAN RTX). #Params counts the trainable parameters. Δ columns show the relative change with respect to the standard Baseline.

Setting	Method	N	#Params	Time (min)	Δ Time	Memory (GiB)	Δ Memory
RGB 20%	Baseline	30,720	368,640	26.6	—	18.9	—
	Baseline ⁺	33,280	399,360	28.1	+5.7%	20.3	+7.2%
	CVQPG	30,720	399,360	26.5	0.0%	18.9	−0.1%
RGB 10%	Baseline	15,360	184,320	17.0	—	10.9	—
	Baseline ⁺	16,640	199,680	17.8	+4.5%	11.6	+5.7%
	CVQPG	15,360	199,680	17.0	0.0%	10.9	−0.1%
Grayscale 20%	Baseline	15,360	122,880	13.0	—	9.0	—
	Baseline ⁺	17,280	138,240	14.1	+8.2%	9.9	+10.3%
	CVQPG	15,360	138,240	13.4	+3.0%	9.5	+5.9%
Grayscale 10%	Baseline	7,680	61,440	8.4	—	5.2	—
	Baseline ⁺	8,640	69,120	9.0	+7.1%	5.7	+9.4%
	CVQPG	7,680	69,120	8.7	+3.4%	5.5	+5.2%

3 Per-Scene Results

Tab. 2 reports the results of all three models on every scene and setting. CVQPG improves upon the Baseline in all 200 comparisons and upon the Baseline⁺ in 182 of them. The results show that CVQPG provides consistent improvements at equal primitive counts, and these gains are unbiased across the dataset.

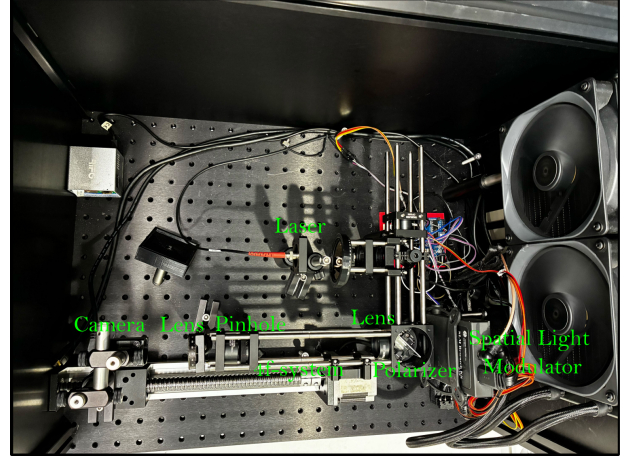


Figure 1: Holographic display prototype in the lab.

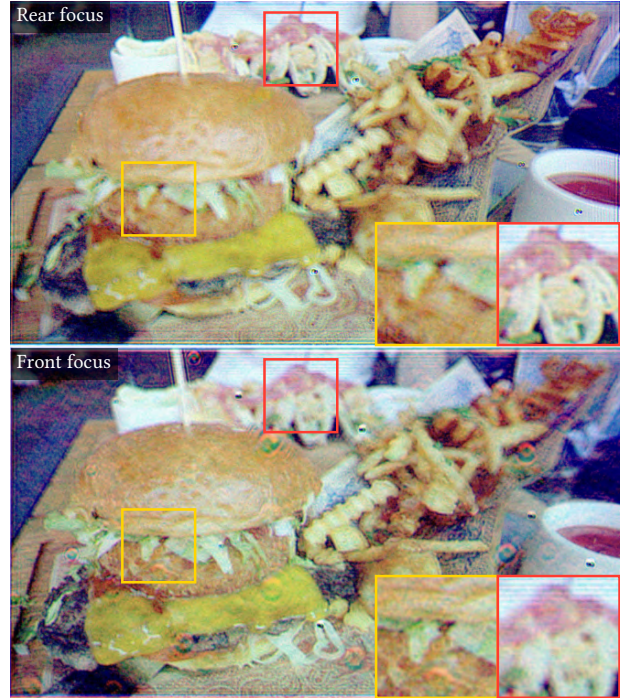


Figure 2: Experimentally captured reconstruction result of the Burger scene on our prototype. We display phase-only holograms encoded from the CVQPG representation and photograph the reconstruction at each depth plane. (Source Image: kennejima)

4 Visual Results

Fig. 3 illustrates two representative reconstructions from our dataset.

Table 2: Per-scene results for all four settings, averaged over the two depth planes. Each cell lists Baseline / Baseline⁺ / CVQPG. Our approach CVQPG is the best of the three in 182 of the 200 cells.

Scene	Rate	RGB					Grayscale				
		PSNR↑	SSIM↑	LPIPS↓	FLIP↓	CVVDP↑	PSNR↑	SSIM↑	LPIPS↓	FLIP↓	CVVDP↑
Pencil	20%	26.967/27.094/27.115	.7737/.7766/.7792	.2946/.2893/.2876	.1528/.1516/.1508	8.749/8.770/8.780	26.788/27.032/27.074	.7738/.7820/.7861	.3829/.3783/.3690	.0856/.0826/.0831	8.592/8.673/8.683
	10%	25.798/25.935/25.931	.7391/.7441/.7434	.3295/.3284/.3239	.1700/.1660/.1684	8.488/8.534/8.520	25.364/25.602/25.557	.7319/.7365/.7403	.4268/.4203/.4118	.1006/.0991/.0982	8.176/8.243/8.256
Sushi	20%	33.646/33.766/33.904	.9199/.9222/.9242	.2555/.2542/.2504	.0826/.0817/.0804	9.511/9.519/9.536	34.382/34.647/34.843	.9286/.9323/.9355	.3227/.3202/.3132	.0518/.0504/.0492	9.412/9.447/9.483
	10%	32.364/32.527/32.589	.8955/.8987/.9002	.2888/.2848/.2836	.0935/.0916/.0913	9.347/9.371/9.381	32.928/33.122/33.369	.9023/.9077/.9116	.3577/.3506/.3455	.0609/.0589/.0574	9.133/9.186/9.228
Flower	20%	32.554/32.682/32.793	.9105/.9131/.9158	.2469/.2409/.2373	.0843/.0839/.0826	9.540/9.555/9.579	33.111/33.362/33.522	.9190/.9237/.9273	.3204/.3167/.3095	.0642/.0631/.0620	9.417/9.451/9.491
	10%	31.179/31.307/31.413	.8797/.8831/.8858	.2916/.2868/.2834	.0959/.0945/.0939	9.285/9.313/9.340	31.523/31.807/31.945	.8820/.8895/.8946	.3643/.3561/.3477	.0744/.0722/.0714	9.054/9.126/9.174
Straw	20%	32.322/32.457/32.531	.9018/.9041/.9062	.2306/.2282/.2263	.0926/.0919/.0914	9.352/9.376/9.371	32.696/32.976/33.074	.9097/.9143/.9173	.3415/.3354/.3293	.0604/.0589/.0584	9.208/9.254/9.286
	10%	31.082/31.214/31.310	.8777/.8805/.8829	.2640/.2599/.2570	.1011/.1003/.0989	9.243/9.236/9.279	31.023/31.335/31.387	.8770/.8832/.8866	.3760/.3689/.3651	.0716/.0691/.0685	8.826/8.904/8.935
Dragon	20%	27.448/27.560/27.604	.8306/.8337/.8363	.2916/.2865/.2854	.1395/.1389/.1379	9.071/9.093/9.103	27.873/28.134/28.139	.8386/.8435/.8476	.3801/.3773/.3716	.0883/.0866/.0860	8.802/8.849/8.869
	10%	26.152/26.330/26.328	.7952/.8002/.8013	.3273/.3206/.3220	.1566/.1543/.1534	8.810/8.856/8.856	26.282/26.565/26.580	.7922/.8009/.8045	.4182/.4101/.4049	.1051/.1013/.1005	8.375/8.455/8.479
Tiger	20%	28.991/29.096/29.141	.8549/.8568/.8599	.3008/.2967/.2930	.1087/.1081/.1071	9.252/9.271/9.283	29.124/29.332/29.377	.8574/.8625/.8670	.3840/.3801/.3719	.0732/.0715/.0712	9.015/9.069/9.098
	10%	27.968/28.070/28.109	.8272/.8314/.8330	.3391/.3325/.3280	.1205/.1189/.1184	8.996/9.032/9.042	28.027/28.201/28.303	.8278/.8335/.8385	.4165/.4116/.4049	.0830/.0810/.0799	8.693/8.756/8.797
Windmill	20%	24.709/24.840/24.886	.6859/.6902/.6956	.3326/.3290/.3259	.1388/.1371/.1361	8.826/8.851/8.880	24.995/25.244/25.284	.6784/.6874/.6965	.4680/.4604/.4528	.0952/.0920/.0914	8.348/8.414/8.462
	10%	23.441/23.579/23.650	.6450/.6493/.6551	.3702/.3658/.3615	.1590/.1571/.1556	8.509/8.547/8.579	23.785/23.987/24.099	.6339/.6407/.6504	.5032/.4978/.4895	.1107/.1075/.1056	7.909/7.993/8.043
Statue	20%	23.586/23.732/23.749	.6556/.6605/.6652	.3784/.3699/.3701	.1561/.1543/.1533	8.634/8.670/8.709	23.398/23.619/23.729	.6506/.6562/.6677	.5037/.4969/.4867	.1055/.1017/.0993	8.243/8.327/8.393
	10%	22.401/22.538/22.588	.6136/.6184/.6231	.4375/.4304/.4302	.1835/.1800/.1778	8.225/8.279/8.334	22.215/22.361/22.454	.6022/.6096/.6180	.5598/.5513/.5428	.1285/.1248/.1219	7.669/7.767/7.830
Burger	20%	32.860/32.940/33.030	.8947/.8966/.8985	.2758/.2736/.2700	.0745/.0740/.0730	9.576/9.583/9.600	32.967/33.185/33.306	.8993/.9035/.9070	.3531/.3494/.3447	.0545/.0528/.0521	9.413/9.450/9.482
	10%	31.738/31.890/31.931	.8679/.8716/.8730	.3106/.3074/.3014	.0843/.0824/.0826	9.358/9.393/9.403	31.687/31.889/32.044	.8689/.8746/.8794	.3794/.3724/.3668	.0636/.0615/.0603	9.101/9.168/9.214
Bento	20%	29.567/29.693/29.746	.8797/.8822/.8832	.2879/.2850/.2823	.1142/.1125/.1123	9.166/9.168/9.189	29.703/29.997/29.995	.8841/.8886/.8908	.3567/.3513/.3453	.0746/.0734/.0720	8.899/8.946/8.947
	10%	28.218/28.403/28.401	.8515/.8570/.8549	.3336/.3243/.3257	.1275/.1255/.1252	8.907/8.935/8.926	27.925/28.302/28.242	.8463/.8536/.8552	.4025/.3953/.3901	.0897/.0862/.0862	8.545/8.614/8.620
Mean	20%	29.265/29.386/29.450	.8307/.8336/.8364	.2895/.2853/.2828	.1144/.1134/.1125	9.168/9.186/9.203	29.504/29.753/29.834	.8340/.8394/.8443	.3813/.3766/.3694	.0753/.0733/.0725	8.935/8.988/9.019
	10%	28.034/28.179/28.225	.7992/.8034/.8053	.3292/.3241/.3217	.1292/.1271/.1265	8.917/8.950/8.966	28.076/28.317/28.398	.7964/.8030/.8079	.4204/.4134/.4069	.0888/.0862/.0850	8.548/8.621/8.658

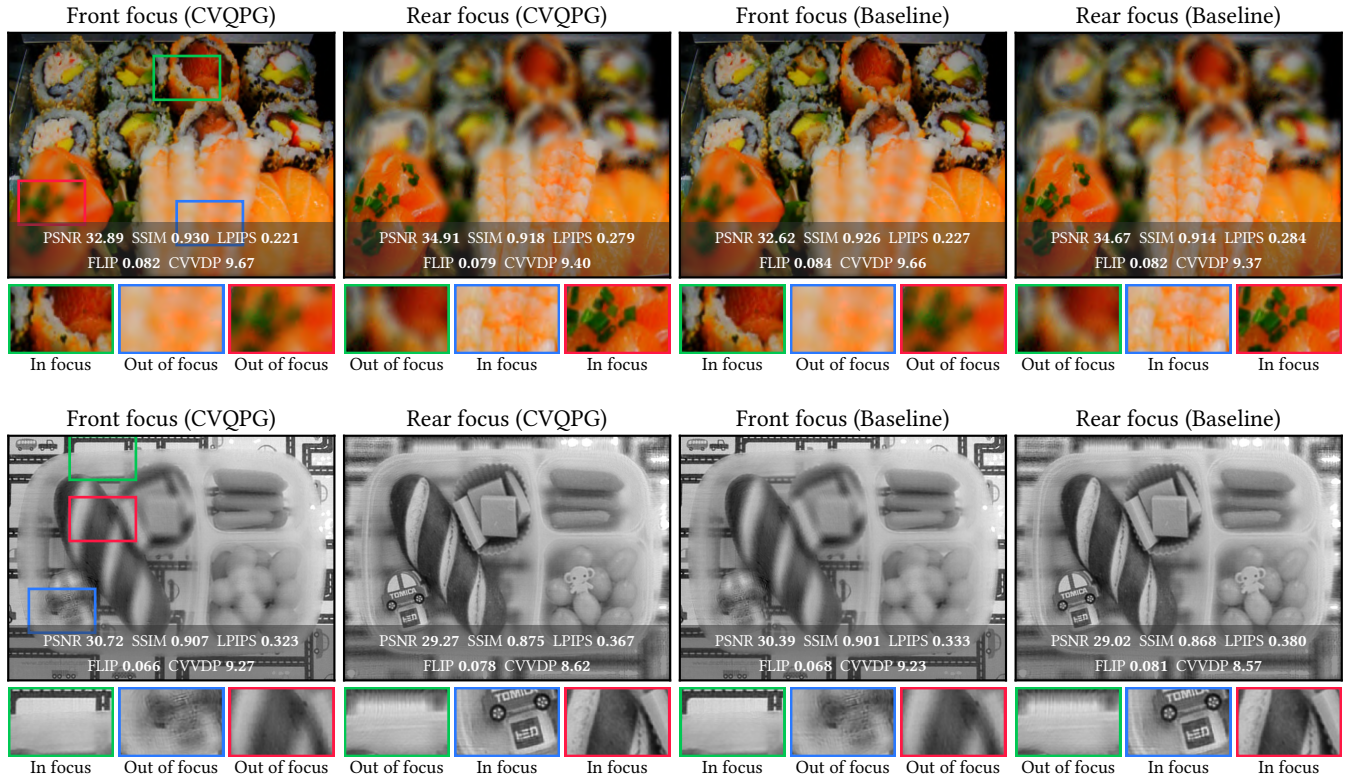


Figure 3: Visual comparison on simulated holographic reconstructions between CVQPG and the baseline. We list two representative scenes of RGB 20% and grayscale 20%. (Source Image: Ben Sutherland, anotherlunch.com)